

Victoria Park Academy Shenzhen

CIE IGCSE Computer Science (0478) — YEAR 9 SEMESTER 1

ASSESSMENT MATERIALS & REPORTING PACK

This pack contains:

1. In-Class Quizzes (Weeks 3, 6, 10, 14)
2. End-of-Semester Student Report Template
3. Class-Level Progress Tracker
4. Bilingual Glossary Poster (A4 Printable)

PART 1: IN-CLASS QUIZZES

QUIZ 1 — BINARY & HEX (WEEK 3)

Name: _____ Class: _____ Date: _____

Time: 15 minutes | Total Marks: 20

SECTION A: Multiple Choice (5 marks)

Circle the correct answer.

1. What is the denary value of 00001111?
A) 7 B) 15 C) 16 D) 31
2. What is 255 in hexadecimal?
A) FE B) FF C) 100 D) 256
3. Which place value is represented by the leftmost bit in 8-bit binary?
A) 1 B) 64 C) 128 D) 256
4. What is the binary result of 00001010 + 00000101?
A) 00001111 B) 00010000 C) 00001011 D) 00010101
5. How many bits are in one byte?
A) 4 B) 8 C) 16 D) 32

SECTION B: Short Answer (10 marks)

Show your working.

10.6. Convert 100 to 8-bit binary. [2 marks]

11.7. Convert 1010 1100 to denary. [2 marks]

12.8. Convert 3E (hex) to denary. [2 marks]

13.9. Explain what is meant by overflow in 8-bit binary. Give an example. [2 marks]

14. 10. Why do computer scientists use hexadecimal instead of binary? [2 marks]

SECTION C: Problem Solving (5 marks)

15. 11. A computer stores student ages in 8 bits.

(a) What is the oldest age possible? [1 mark] _____

(b) Is this enough for all ages? Explain. [2 marks]

(c) How many bits would be needed to store age 500? [2 marks]

ANSWER KEY — QUIZ 1 (BINARY & HEX)

SECTION A: 1. B (15) 2. B (FF) 3. C (128) 4. A (00001111 = 15) 5. B (8)

SECTION B: 6. 01100100 (64 + 32 + 4 = 100) [2 marks: 1 correct, 1 working]

7. 172 (128 + 32 + 8 + 4 = 172) [2 marks: 1 correct, 1 working]

8. 62 (3 x 16 + 14 = 62) [2 marks: 1 correct, 1 working]

9. Overflow occurs when result exceeds 255 (max for 8 bits) [1];

example: 200 + 100 = 300 [1]

10. Hex is shorter/easier to read than long binary strings [1];

one hex digit = 4 bits (one nibble) [1]

SECTION C: 11. (a) 255 [1]

(b) No, some people live longer than 255 [1]; need more bits [1]

(c) 9 bits ($2^8 = 256$, $2^9 = 512$; $500 < 512$) [2]

QUIZ 2 — DATA REPRESENTATION TEST (WEEK 6)

Name: _____ **Class:** _____ **Date:** _____

Time: 30 minutes | Total Marks: 35

SECTION A: Multiple Choice (5 marks)

16. 1. How many bits are needed to store one ASCII character?

A) 4 B) 7 C) 8 D) 16

17. 2. What is the main difference between ASCII and Unicode?

A) ASCII is faster

B) Unicode supports more characters including Chinese

C) Unicode uses fewer bits

D) ASCII is newer

18. 3. An image has a resolution of 640 x 480 with 16-bit colour depth.

How many bits are needed to store the uncompressed image?

A) 307,200 B) 4,915,200 C) 614,400 D) 1,228,800

19. 4. Which is an example of lossy compression?

A) PNG B) GIF C) JPEG D) BMP

20. 5. What happens during sound sampling?

A) The sound is made louder

B) The analogue wave is measured at regular intervals

C) The sound is converted to text

D) The file is compressed

SECTION B: Short Answer (15 marks)

21. 6. Explain why Unicode is needed for representing text in Chinese. [2 marks]

22. 7. A text file contains 1000 characters using ASCII encoding.

Calculate the file size in bytes. [2 marks]

23. 8. An image is 1024 x 768 pixels with 24-bit colour depth.

(a) Calculate the file size in bytes. Show your working. [3 marks]

(b) If the image is compressed using lossy compression to 500KB,

explain what happens to the quality. [2 marks]

24. 9. A sound file is recorded at 44.1 kHz with 16-bit samples and 2 channels (stereo).

Calculate the file size for 1 minute of uncompressed audio in bytes. [3 marks]

25. 10. Explain the difference between lossy and lossless compression. [3 marks]

SECTION C: Extended Answer (15 marks)

26. 11. Victoria Park Academy Shenzhen wants to store digital resources for students.

They have three types of files: text documents, photographs, and audio recordings.

(a) For each file type, explain the best compression method and why. [6 marks]

Text: _____

Photographs: _____

Audio: _____

(b) Explain why metadata is important for image files. [3 marks]

(c) A student says: "Lossy compression is always bad because it loses data."

Evaluate this statement. [3 marks]

(d) Calculate how many Chinese characters can be stored in 2KB using Unicode (16-bit). [3 marks]

ANSWER KEY — QUIZ 2 (DATA REPRESENTATION TEST)

SECTION A: 1. B (7 bits, though often stored in 8) 2. B 3. B (640 x 480 x 16 = 4,915,200)

4. C (JPEG) 5. B

SECTION B: 6. ASCII only supports 128 characters (English) [1]; Unicode supports thousands including Chinese characters [1]

7. 1000 characters x 1 byte = 1000 bytes [2]

8. (a) $1024 \times 768 \times 24 = 18,874,368$ bits [1]; $/ 8 = 2,359,296$ bytes [1]; working shown [1]

(b) Some image data is permanently lost [1]; quality decreases but file is smaller [1]

9. $44,100 \times 16 \times 2 \times 60 = 84,672,000$ bits [2]; $/ 8 = 10,584,000$ bytes [1]

10. Lossy = permanent data loss, smaller files, cannot recover original [1.5];

Lossless = no data loss, fully recoverable, larger files [1.5]

SECTION C: 11. (a) Text: Lossless (RLE or dictionary) — text must be fully recoverable [2]

Photographs: Lossy (JPEG) — small quality loss acceptable, much smaller files [2]

Audio: Lossy (MP3) or Lossless (FLAC) — depends on use case [2]

(b) Metadata stores information about the image [1]; camera settings, date, location [1];

helps organise and search files [1]

(c) Not always bad [1]; acceptable for photos/audio where small quality loss is OK [1];

not suitable for text or medical images where accuracy matters [1]

(d) 2KB = 2048 bytes = 16,384 bits [1]; $16,384 / 16 = 1024$ characters [2]

QUIZ 3 — HARDWARE (WEEK 10)

Name: _____ Class: _____ Date: _____

Time: 15 minutes | Total Marks: 20

SECTION A: Multiple Choice (5 marks)

27. 1. Which register holds the address of the next instruction to be fetched?

- A) ACC B) MAR C) PC D) MDR

28. 2. What happens to data in RAM when the computer is turned off?

- A) It is saved to the hard drive
B) It is lost
C) It is stored in cache
D) Nothing happens

29. 3. Which memory type is fastest?

- A) RAM B) Cache C) Virtual memory D) Hard disk

30. 4. What is the purpose of the ALU in the CPU?

- A) To store data
B) To perform arithmetic and logic operations
C) To control the flow of data
D) To fetch instructions

31. 5. Which storage device has no moving parts?

- A) HDD B) CD/DVD C) SSD D) Magnetic tape

SECTION B: Short Answer (10 marks)

32. 6. Describe the fetch-decode-execute cycle. [3 marks]

33.7. Explain the difference between RAM and ROM. [2 marks]

34.8. Why is cache memory faster than RAM? [2 marks]

35.9. What is virtual memory and why is it slower than RAM? [3 marks]

SECTION C: Problem Solving (5 marks)

36.10. A computer has 4GB RAM and a student opens:

- 10 Chrome tabs (200MB each)
- Photoshop (2GB)
- Spotify (100MB)
- Word (300MB)

(a) Calculate the total memory needed. [2 marks]

(b) What will the operating system do when RAM is full? [2 marks]

(c) How could the student improve performance without buying more RAM? [1 mark]

ANSWER KEY — QUIZ 3 (HARDWARE)

SECTION A: 1. C (PC) 2. B (It is lost) 3. B (Cache) 4. B (Arithmetic and logic)

5. C (SSD)

SECTION B: 6. Fetch: PC → MAR → Memory → MDR → CIR; PC increments [1];

Decode: CU interprets instruction [1];

Execute: ALU performs operation [1]

7. RAM is volatile/read-write [1]; ROM is non-volatile/read-only [1]

8. Cache is closer to CPU [1]; uses faster memory technology [1]

9. Virtual memory uses hard disk as RAM [1]; hard disk is much slower than RAM [1];
causes thrashing if overused [1]

SECTION C: 10. (a) $10 \times 200 + 2000 + 100 + 300 = 4600\text{MB} = 4.6\text{GB}$ [2]

(b) Uses virtual memory (hard disk) [1]; computer slows down [1]

(c) Close unused tabs/programs [1]

QUIZ 4 — SOFTWARE & NETWORKS (WEEK 14)

Name: _____ Class: _____ Date: _____

Time: 15 minutes | Total Marks: 20

SECTION A: Multiple Choice (5 marks)

37. 1. What is the main function of an operating system?

- A) To write programs
- B) To manage computer resources
- C) To browse the internet
- D) To design websites

38. 2. Which translator works line by line?

- A) Compiler
- B) Interpreter
- C) Assembler
- D) Linker

39. 3. What does DNS do?

- A) Encrypts data
- B) Translates domain names to IP addresses
- C) Blocks viruses
- D) Manages files

40. 4. What does the 'S' in HTTPS stand for?

- A) Simple
- B) Secure
- C) Standard
- D) System

41. 5. Which is NOT a type of malware?

- A) Firewall
- B) Trojan
- C) Worm
- D) Ransomware

SECTION B: Short Answer (10 marks)

42. 6. Explain the difference between a compiler and an interpreter. [2 marks]

43. 7. Describe the role of DNS when you visit a website. [3 marks]

44. 8. Explain why HTTPS is more secure than HTTP. [2 marks]

45. 9. Describe two cybersecurity threats and how to protect against each. [3 marks]

SECTION C: Scenario (5 marks)

46. 10. Victoria Park Academy Shenzhen is setting up a new computer network.

(a) Explain why packet switching is better than circuit switching for the school. [2 marks]

(b) Describe two cybersecurity measures the school should implement. [3 marks]

ANSWER KEY — QUIZ 4 (SOFTWARE & NETWORKS)

SECTION A: 1. B (Manage resources) 2. B (Interpreter) 3. B (Translates domain names to IP)

4. B (Secure) 5. A (Firewall is a protection, not malware)

SECTION B: 6. Compiler translates entire program at once [1]; interpreter translates line by line [1]

7. Browser sends domain name to DNS server [1]; DNS returns IP address [1];

browser uses IP to contact web server [1]

8. HTTPS encrypts data using SSL/TLS [1]; prevents eavesdropping and protects passwords [1]

9. Any two threats with matching protection [3 marks total]

e.g., Phishing → email filters, education; Malware → antivirus, updates

SECTION C: 10. (a) More efficient bandwidth use [1]; multiple users can share network [1]

(b) Any two: firewall, antivirus, 2FA, encryption, backups, updates [3 marks]

PART 2: END-OF-SEMESTER STUDENT REPORT TEMPLATE

VICTORIA PARK ACADEMY SHENZHEN

CIE IGCSE COMPUTER SCIENCE (0478) — YEAR 9 SEMESTER 1

STUDENT PROGRESS REPORT

Student Name: _____ Class: _____

Teacher: _____ Date: _____

ATTAINMENT SUMMARY

Assessment	Score	Max	%	Grade
-----	-----	-----	---	-----
Binary/Hex Quiz (Wk 3)			20	
Data Representation Test (Wk 6)			35	
Hardware Quiz (Wk 10)			20	
Software & Networks Quiz (Wk 14)			20	
Mock Exam (Wk 18)			35	
TOTAL			130	

Grade Boundaries: A* = 80%+, A = 70%+, B = 60%+, C = 50%+, D = 40%+

TOPIC BREAKDOWN

Rate student performance for each topic (1 = Emerging, 2 = Developing, 3 = Secure):

Topic	Wk	Rating	Evidence
-----	----	-----	-----
Binary & Denary	1-2		
Binary Addition & Overflow	2		
Hexadecimal	3		
Text Representation (ASCII/Unicode)	4		
Images & Resolution	5		
Sound & Compression	6		
Packet Switching	7		
Error Detection & Encryption	8		
CPU & FDE Cycle	9		
Memory (RAM/ROM/Cache)	10		
Storage & I/O Devices	11		
Operating Systems	12		
Translators & IDEs	12		
Internet, URLs & DNS	13		
HTTP/HTTPS & IP Addresses	13		
Cybersecurity Threats	14		
Cybersecurity Protections	14		
Digital Currency & Blockchain	15		
AI, Machine Learning & Robotics	15		
Python Basics	16		

STRENGTHS

47.1. _____
48.2. _____
49.3. _____

AREAS FOR DEVELOPMENT

50.1. _____
51.2. _____
52.3. _____

TARGETS FOR YEAR 10

53.1. _____
54.2. _____
55.3. _____

TEACHER COMMENTS

STUDENT REFLECTION

I agree / disagree with this assessment because:

One thing I will do to improve:

Student Signature: _____ Date: _____

Parent/Guardian Signature: _____ Date: _____

PART 3: CLASS-LEVEL PROGRESS TRACKER

VICTORIA PARK ACADEMY SHENZHEN

CIE IGCSE COMPUTER SCIENCE (0478) — YEAR 9 SEMESTER 1

CLASS PROGRESS TRACKER

Class: _____ Teacher: _____ Date: _____

STUDENT ATTAINMENT OVERVIEW

| Student Name | Q1 (20) | Q2 (35) | Q3 (20) | Q4 (20) | Mock (35) | Total (130) | % | Grade
|

|-----|-----|-----|-----|-----|-----|-----|---|-----|

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| CLASS AVERAGE | | | | | | | |

TOPIC MASTERY MATRIX (Class Level)

For each topic, record number of students at each level:

E = Emerging, D = Developing, S = Secure

| Topic | E | D | S | % Secure | Action Needed |

|-----|---|---|---|-----|-----|

| Binary & Denary | | | | |

| Binary Addition | | | | |

| Hexadecimal | | | | |

| Text Representation | | | | |

| Images | | | | |

| Sound & Compression | | | | |

| Packet Switching | | | | |

| Error Detection | | | | |

| CPU & FDE Cycle | | | | |

| Memory | | | | |

| Storage & I/O | | | | |

| Operating Systems | | | | |

| Translators & IDEs | | | | |

| Internet & URLs | | | | |

| HTTP/HTTPS | | | | |

| Cybersecurity Threats | | | | |

| Cybersecurity Protections | | | | |

| Digital Currency | | | | |

| Blockchain | | | | |

| AI & Machine Learning | | | | |

| Python Basics | | | | |

| Variables & I/O | | | | |

CLASS GAP ANALYSIS

TOP 3 CLASS STRENGTHS:

56.1. _____

57.2. _____

58.3. _____

TOP 3 AREAS FOR IMPROVEMENT:

59.1. _____

60.2. _____

61.3. _____

INTERVENTIONS PLANNED:

SEMESTER REFLECTION

What worked well this semester?

What will I change for next semester?

PART 4: BILINGUAL GLOSSARY POSTER (A4 PRINTABLE)

VPA COMPUTER SCIENCE — BILINGUAL KEYWORD GLOSSARY

UNIT 1: DATA REPRESENTATION

binary	二进制	er jin zhi
denary/decimal	十进制	shi jin zhi
hexadecimal	十六进制	shi liu jin zhi
bit	位	wei
byte	字节	zi jie
place value	位值	wei zhi
ASCII	美国信息交换标准代码	一
Unicode	统一码	tong yi ma
pixel	像素	xiang su
resolution	分辨率	fen bian lv
colour depth	颜色深度	yan se shen du
compression	压缩	ya suo
lossy	有损压缩	you sun ya suo
lossless	无损压缩	wu sun ya suo

UNIT 2: DATA TRANSMISSION

packet switching	分组交换	fen zu jiao huan
data packet	数据包	shu ju bao
header	报头	bao tou
error detection	错误检测	cuo wu jian ce
parity	奇偶校验	ji ou jiao yan
encryption	加密	jia mi

UNIT 3: HARDWARE

CPU | 中央处理器 | zhong yang chu li qi
fetch-decode-execute | 取指-译码-执行 | qu zhi - yi ma - zhi xing
register | 寄存器 | ji cun qi
program counter | 程序计数器 | cheng xu ji shu qi
RAM | 随机存取存储器 | sui ji cun qu cun chu qi
ROM | 只读存储器 | zhi du cun chu qi
cache | 高速缓存 | gao su huan cun
virtual memory | 虚拟内存 | xu ni nei cun
sensor | 传感器 | chuan gan qi
embedded system | 嵌入式系统 | qian ru shi xi tong

UNIT 4: SOFTWARE

operating system | 操作系统 | cao zuo xi tong
compiler | 编译器 | bian yi qi
interpreter | 解释器 | jie shi qi
IDE | 集成开发环境 | ji cheng kai fa huan jing

UNIT 5: INTERNET & SECURITY

World Wide Web | 万维网 | wan wei wang
URL | 统一资源定位符 | tong yi zi yuan ding wei fu
DNS | 域名系统 | yu ming xi tong
HTTP | 超文本传输协议 | chao wen ben chuan shu xie yi
HTTPS | 安全超文本传输协议 | an quan chao wen ben chuan shu xie yi
IP address | IP 地址 | IP di zhi
router | 路由器 | lu you qi
firewall | 防火墙 | fang huo qiang
malware | 恶意软件 | e yi ruan jian
phishing | 网络钓鱼 | wang luo diao yu
two-factor auth | 双因素认证 | shuang yin su ren zheng

UNIT 6: EMERGING TECH

artificial intelligence | 人工智能 | ren gong zhi neng
machine learning | 机器学习 | ji qi xue xi
robotics | 机器人技术 | ji qi ren ji shu
blockchain | 区块链 | qu kuai lian
digital currency | 数字货币 | shu zi huo bi

UNIT 7: PROGRAMMING

Python	Python	Python
syntax	语法	yu fa
variable	变量	bian liang
input	输入	shu ru
output	输出	shu chu
string	字符串	zi fu chuan
integer	整数	zheng shu
comment	注释	zhu shi

END OF ASSESSMENT & REPORTING PACK